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Comparing stylistic variance in the Pauline and contemporary Greek corpora

Running Title (<50 characters)

Pauline vs. contemporary stylistic variance

Abstract (<200 words)

Two persistent problems plague Pauline stylometry: (1) the absence of a verifiably authentic Pauline reference corpus, and (2) the resulting inability to select optimal discriminating features. Previous research presupposes which letters are authentic, relies on pre-selected feature lists, analyses single features in isolation, or operates in high-dimensional feature space. This paper proposes feature selection through probabilistic comparison with contemporaneous Greek corpora scholarship treats as univocal, operationalising style through computational feature discovery across authors dated 100 BC to AD 100. Method 1 evaluates corpus-level distinctiveness and tests robustness against Plato. Method 2 assesses attributive power at the individual-text level, using classical NLP tooling and transformer-based embedding models. The Pauline corpus displays high internal variance, exceeding many multi-text authors, yet falls within a plausible range of univocality against contemporaneous comparanda. Method 2 finds no feature above 50.5% on 32-class attribution, an aggregate driven largely by the single-text-author bloc; transformer embedding dimensions contribute no discriminative signal beyond the classical character bigram $\varsigma\alpha$ once the scale floor is properly calibrated. The Pauline letters are stylometrically heterogeneous, yet remain within bounds observed for other univocal ancient Greek corpora, recommending probabilistic rather than binary framings of computationally derived authorship judgments.

Keywords (≤ 10)

Paul, Plato, Stylometry, Digital Humanities, Authorship attribution, Koine Greek, New Testament, Authenticity

Background

Computational stylometric analyses aimed at discriminating between authentic and inauthentic letters of the Pauline corpus have predominantly focused on single stylistic variables at a time, such as White,[1] or on a curated list of variables selected by the study authors, such as Pracht and McCauley.[2] This involved approach to feature selection is problematic for several reasons.

First, while it is possible to discriminate authors with individual features,[3] there are no definitively authentic Pauline letters from which to define a baseline. When study authors curate individual features or lists of features, they are making an *a priori* judgement about which features are authentically Pauline, and the only source for those features is the Pauline letters themselves. The authentic Pauline baseline is therefore asserted, not discovered. Comparisons are then made which inevitably produce discoveries of difference (as do all statistical comparisons of non-identical texts), which then lead to judgments about the degree

of intra-textual variation between the studied letter and the proclaimed baseline. A. Q. Morton describes the issue: "If you ask them [scholars] why they consider them [epistles] to be Pauline they can only say that they see in them the mind and style of Paul. If you ask them how they know the mind and style of Paul they can only reply that they see it in his epistles. The argument is circular." [4] Morton himself axiomatically accepted Galatians as authentic, leading him to acknowledge that "When I talk of Paul as the author of 1st Corinthians I mean that 1st Corinthians was written by the man who wrote Galatians." [5] This replicates the circular argument: Morton presupposes Pauline authenticity of a letter when authenticity is the subject of the question.

It also means that the accuracy of the chosen features is unevaluable. Whether or not a chosen feature discriminates Paul correctly is unknowable, since which letters of Paul are authentic is itself unknowable. A second layer of circularity is thereby introduced. How can one determine whether the selected features are useful? Only by analysing the downstream results. And how are the downstream results analysed? Only by referring back to the predetermined authentic corpus.

Paul's individual level of career-wide stylistic variation is also unknowable. If multiple authors are involved in the production of any of the letters, it is impossible to statistically disentangle their contributions from Paul's. Errors introduced from numerical author uncertainty have been demonstrated in clustering analyses of the American *Federalist Papers*, where additional hypothetical authors led to widespread model inaccuracies. [6] In statistical terms, we end up dealing with an unprovable premise.

Since pre-selected feature curation, when dealing with unverifiable corpora, does not produce inductively discovered authorial style and tends to produce biased results, a logical goal is to minimise bias in the feature selection process itself. Bias cannot be removed wholesale, but methods that "[relieve] the user from critical design choices (e.g., feature set)" offer improvements. [7] All features should be available to the study, while "the method itself finds the optimal combination for the problem at hand." [8] Authentic letters should not be presumed, available features should not be discarded before runtime, and features should be assessed on the evidential strength of stylistic results.

A second issue follows: there are no known universal discriminating features and no feature-based definitions of authorial style, so there is a gap in objectively assessing the evidential strength of results. A feature's discriminating power changes depending on the corpora to which it is applied, which rules out any appeal to an objective discriminator, and there is no agreed list of features and ranges that agglomerates into a concrete concept called "style."

Faced with these challenges, one might consider measuring as many features as possible and applying a reduction technique to lower the noise floor. Such approaches have been empirically shown to generalise poorly to unrelated corpora when applied to the Pauline corpus. [9] This is an instance of the peaking, or Hughes, phenomenon, "which in its simplest manifestation takes the form of decreasing expected error following [sic] by increasing expected error for increasingly large feature sets. Given a sequence, $x_1; x_2; \dots; x_d; \dots$, of features, at first there is decrease in expected error as d increases and then, after some point, an increase in error for increasing d ." [10] An objective optimal number of features therefore exists, but identifying it without a verifiable authentic reference corpus seems unworkable.

The two core issues in Pauline stylometrics can thus be summarised as (1) the absence of a verifiably authentic Pauline reference corpus, which causes (2) the inability to select optimal discriminating features. These problems present serious challenges for Pauline

stylometric studies. The approach proposed here does not dissolve the circularity. It replaces the internal, self-referential reference frame — Paul's putative authentic core — with an external empirical one: the contemporaneous Koine authors whom scholarship regards as univocal. The internal frame asks how much a given letter deviates from the presumed-authentic Paul, a question which is unanswerable because its baseline is asserted rather than discovered. The external frame asks whether Paul's internal variance falls within the range observed across other authors of the period. That question is answerable, and the answer is falsifiable against data drawn from outside the Pauline corpus. In what follows we propose two new methods of computational analysis for the Pauline corpus focused on feature discovery through contemporaneous texts.

A (highly) compressed history of Pauline and Platonic stylometrics

Pre-computational stylometrics faced an obvious limitation: texts had to be processed manually, so scholars ended up with small datasets (small relative to computational corpora) and lacked the conditions required for uncovering textual features in any large sample. Features are broadly defined as quantifiable textual elements by which the analyst can "uncover the 'characteristics' of an author", assisting in questions of authorship attribution and chronology.[11] Feature selection, as a result of the manual nature of data collection, was limited: analyses initially centred on various lexical distributions (e.g. the frequency of $\mu\eta\nu$ per page). Investigated features were therefore either in low frequency or "isolated", as in reply formulae. Some scholars, such as Lutoslawski, did devise larger feature sets, but their usefulness was curbed by arbitrary evaluation of feature importance.[12] Chronologically this phase of scholarship runs from roughly the mid-nineteenth century to the mid-twentieth.

The overarching theme for Platonic studies in this period, in terms of authorship attribution by content-based argumentation and early stylometrical analyses, can be generalised: "[i]n the nineteenth century it was fashionable to reject a still greater number [of dialogues], reducing the total of accepted dialogues to a dozen or less".[13] Something similar happened in Pauline studies. From the initial opening of the authenticity question in the eighteenth century, much of the nineteenth century (and much of the twentieth) was occupied with the scholarly tug-of-war between the voice of "tradition", which found all fourteen letters authentic, and theories increasingly concluding more letters to be inauthentic: Baur's, which treated four *Hauptbriefe* as authentic, or that of the Radical Dutch school, which proclaimed the whole Pauline corpus pseudepigraphal.[14] These conclusions were not, of course, the broad consensus of scholarship, but they substantiate a general *tendency* in assigning authorship within unverifiable corpora: the rejection of writings containing features perceived as discordant with what was held to be the key writings of a given author.

The advent of stylometric analyses within Pauline studies belongs to the same period, although the methodological nomenclature varies drastically. Schleiermacher notably claimed 1 Timothy as not authentically Pauline, and the distribution of *hapax legomena* played a central role in these early approaches to the Pauline stylome alongside other linguistic factors. Scholars like Harrison concluded to an identifiable chronological development of Pauline style which they argued could not be reconciled with the Pastorals.[15] Harrison is a telling example of the period: he confidently states that "[t]his figure includes of course a large number of those commonest nouns, verbs, prepositions, &c. [...] neither singly nor

collectively does the presence weigh so much as dust in the balance in favour of the Pauline authorship",[16] and his diagrams, in which the progression of *hapax legomena* and other lexical distributions is observed from the main Pauline group (Ro, 1–2 Co, Gal, Phl, Phm, 1–2 Th & Col) to the Pastorals (1–2 Ti & Tit), are mildly misleading owing to the constant jumbling of the position of individual letters across different diagrams. The effect is to suggest a chronological development of the Pauline stylome that the underlying data do not support.[17] Harrison is an outlier in methodological recklessness and statistical thoroughness, yet the basic dismissal of frequent lexical elements in the definition of a stylome is symptomatic of the era, reinforced by the manual procurement of data.

Conversely, this same era offers prefigurations of later data-rich analyses in the computational age. Ritter's set of forty-three linguistic features, one of the hallmarks of early Platonic stylometry,[18] is a good example. Its central strength consisted in the selection of a larger feature set, consistently applied across the Platonic writings. When judging variance in authorship, however, the basic problem remained the persistent assumption which Ledger opposes: that a work which appears spurious and varied as compared to a "norm" is a sign of different authorship. In the age before computational data collection, the manual collation of even the narrow feature set within the *Corpus Platonicum* was an enormous task, which prevented Ritter from comparing the behaviour of features across other authors. He was, however, troubled by the lack of observed features in Plato's *Menexenus*. [19] It is here that the ability to compute larger amounts of data becomes important.

In the application of computational analysis to ancient texts, Pauline studies were among the forefront. This is feasible due to the existence of biblical concordances combined with the small corpus size of the ancient Greek Bible, which made it convertible to punch cards.[20] A. Q. Morton was the forefather of computational stylometry applied to the New Testament and concentrated his efforts on lexemes of high frequency (typically function words). His enumeration and digitisation of the Pauline texts (and sampled classical *comparanda*) allowed for an evolution: instead of Harrison's examination of *hapax legomena*, Morton focused on the occurrence and positioning of καί, δέ, ἐν, αὐτός and εἶναι, combined with sentence-length distributions, from which he concluded that only the four *Hauptbriefe* (Ro, 1–2 Co & Gal) could be considered authentic.[21] Stylometry thus appeared to corroborate the earlier conclusion reached by Baur, given the close clustering of Morton's selected features in these epistles.

Morton's seemingly straightforward conclusion is shadowed by a considerable caveat: the question of how to judge variance. Where Morton asserts the coherence of the *Hauptbriefe*, David Mealand points out that within individual features (such as the incidence of δέ as the second word in a sentence) there are hints of internal variance that cannot be exclusively explained by a difference in authorship.[22] Contrary to the theory of coherence, Mealand's test of incidence finds a group of seven letters (*Hauptbriefe* + Phl & 1–2 Th) to show less diversity (14.33 for 6 d.f. → *p* below 2.5% but above 5%) than either the *Hauptbriefe* (11.27 for 3 d.f. → *p* below 0.025 and almost 0.01) or a group of ten letters (the seven-letter group + Eph, Col & 2 Ti, yielding 33.63 for 9 d.f. → *p* below 0.001).[23] Probing individual features, then, reveals variance within Morton's supposedly coherent group, which suggests a broader group of coherent letters rather than the extreme inauthenticity verdict of most of the *Corpus Paulinum*. Where Morton is willing to claim inauthenticity with confidence, Mealand calls for caution.[24]

With the development of computational power and the emergence of digitalised corpora of ancient texts, scholars moved beyond function-word counts and hand-collocated *hapax legomena* toward high-dimensional feature spaces, n-gram distributions, character-

level signatures, and, most recently, the internal states of transformer-based language models pretrained on Koine Greek. Pauline studies in the last decade have produced work that is both methodologically more ambitious and epistemologically more restrained. Rosa has cautioned that high-dimensional classifiers trained on a presumed authentic core do not generalise convincingly to unrelated corpora, so that apparent discriminative success is often an artefact of the peaking phenomenon rather than a reliable signal of authorship.[25] Pracht and McCauley, working with a curated feature list, have demonstrated how sensitive inferential conclusions remain to the choice of statistical model even when the underlying counts are held constant.[26] What the recent wave has shown is that computational sophistication has not dissolved the inherited methodological problem of a missing univocal Pauline reference; if anything it has made that absence more visible.

A parallel trajectory is observable in Platonic stylometry, and it is instructive to compare the two fields side by side. Brandwood's survey of stylometry and Platonic chronology traces the maturation of the field from Lutoslawski's arbitrary weighting of features toward more principled, statistically-grounded feature selection, in which the discriminative weight of a given marker is derived from its observed behaviour across the corpus rather than assigned by the analyst's prior intuition.[27] This is the same epistemic movement undergone by Pauline studies from Harrison to Morton and, latterly, to Mealand: a gradual recognition that the features themselves cannot be asked to carry the evidential weight the pre-computational tradition demanded of them.

The clearest illustration of the interpretive split within contemporary Platonic stylometry can be found by juxtaposing the recent work of Jordan Perry and Thomas Koentges. Perry, applying deep learning to the question of the Seventh Epistle's authenticity, is cautious about what a stylistic verdict can deliver. He observes that "[e]xaminations based on the letter's style, however, are subject to the opposite handicap — as shown by the work on the Federalist Papers, stylistic agreement can be seen as decisive evidence towards Platonic authorship, while, given the genre difference between Epistle VII and the Platonic dialogues, a difference in style cannot be attributed solely to different authorship. Essentially, it would be much more difficult to forge an author's subconscious language habits than to ensure concordance with that author's other writings and the general historical record. Thus, content-based examinations should naturally tend to skew towards claiming inauthenticity, while style-based arguments should more often lean towards attempting to show authenticity." [28]

Perry's asymmetry is crucial: stylistic concordance is evidentially stronger than stylistic discordance, because the former is harder to fake than it is to diverge from. Koentges takes the opposite tack in his re-examination of the *Menexenus*, concluding on the basis of a battery of stylometric measures that "there is very little stylistic evidence that the *Menexenus* was written by Plato. The stylometric measures by and large do not cluster the *Menexenus* with other works of the Corpus Platonicum. The *Menexenus* has a higher co-occurrence with Xenophon and Lucian than with Platonic works, and the frequency of καὶ in it is unusual not only when set against other works in the corpus but also when compared with the usage of Plato's Socrates. Furthermore, the genre signal hypothesis was disproved. Taken together, these facts strongly suggest that the *Menexenus* was not written by Plato." [29]

Koentges is willing to advance from stylistic discordance to a positive verdict of inauthenticity; Perry prefers a continuum of authenticity. The methodological difference between them is not one of technique but of inferential charter: what range of variance is one prepared to tolerate within a single authorial stylome before concluding that a different hand is at work? Merely showing stylistic discordance or incoherence is not by itself a warrant for

reassigning authorship. It is a datum which must be weighed against a prior estimate of how much intra-authorial variation one expects to find.

This is the question Ledger raises against the inherited Platonic consensus. He argues against the claim that "atypical works are necessarily spurious", since it is an unwarranted assumption — "for our knowledge of the 'norm' is built on whatever we recognize as genuine"[30] — and this is, in our view, the conceptual anchor for any stylometric programme operating on an unverifiable corpus. The same caution is voiced, from rather different angles, by Pao and van Nes in their respective treatments of the Pastoral Epistles. Both emphasise that the stylistic envelope of a single author can be much wider than the narrow cluster which traditional content-based arguments have been willing to admit.[31] What Ledger, Pao, and van Nes share is a refusal to treat the observed distribution of a feature across a presumed-genuine core as the fixed measure of what is authorially possible. They instead read atypicality as a prompt to revise the estimate of allowable variance rather than as *prima facie* evidence of a different hand.

The same dynamic is played out within Platonic stylometry. Ledger stands alongside Pao and van Nes in underlining the possibility of variance within authorship; Koentges stands closer to Morton, though their methods differ markedly in scope and in scholarly focus.[32] What emerges from this overview is that stylometry is a flexible instrument, heavily dependent on method and data and, above all, interpretation.[33] The approach taken in the present paper seeks to make one thing explicit: the addition of more data through computational means does not, of itself, resolve the methodological problems at the core of the field. It can, however, reframe them. By computing what variance actually looks like across a range of contemporaneous univocal corpora, it becomes possible to ask whether the observed variance within the *Corpus Paulinum* falls within or outside the empirical envelope of Koine Greek authorship in the period between 100 BC and AD 100.

Pitfalls in computational stylometry

Two obvious pitfalls in stylometric method must be addressed at the outset: corruption, and pseudepigrapha (now often called forgeries).[34] The biblical text on which our analysis is based is the *Critical Edition of the Greek New Testament* by the Society of Biblical Literature,[35] an eclectic amalgamation of various manuscripts in which corrupted readings have been, so far as possible, mitigated. The sources for the other ancient texts are (critical) editions supplied by either the Perseus Project or the First1K project. No selected text would be completely satisfactory with regard to corruption and/or interpolation, but there is no available solution to this first concern.

Koentges's second criterion is more difficult to address. The *Corpus Platonicum* is, like the Pauline field, bogged down in discussions of the authenticity of various texts. Koentges criticises Gerard Ledger's early exploration into Platonic stylometry because the question of pseudepigrapha is silently ignored. Dismissal might be an elegant solution to the difficulty within the study of ancient texts, but it is an irresponsible one: a wholly unknown process of transmission and copying has led to the survival of the text, and the question to what extent we can "trust" the final product with which we work is in large part the foundation of any stylometric programme addressed to an ancient corpus. The methodological question is how to frame the analysis of such a field. Do we deem a selection of texts authentic and use that canon as a collection of metrics against which we weigh texts "outside" the canon? Or do we measure variance within an authorial corpus? Ledger *does*

address the possibility of forgery in the *Corpus Platonicum*, either wholesale or through interpolation, but questions the assumption that "atypical works are necessarily spurious and should be excluded from the canon. For our knowledge of the 'norm' is built on whatever we recognize as genuine." [36] We are faced with the same paradox in the selection of texts from which we build any given stylome.

Seen from this angle, and given that Koentges himself conducts an analysis of the *Menexenus* as a possible pseudepigraphal work, the difficulty in both Platonic and Pauline studies is that there is no authorial corpus with one-hundred-percent verifiable authorship. This has been established in Pauline studies since Morton. [37] The disputed *Menexenus* likewise sits within the Platonic canon of tetralogies envisaged by Thrasyllus. [38] The conundrum has the same shape in both fields, and this paper is written with that parallel in view. What more advanced and granular computational analysis can bring to the table is not a dissolution of these difficulties but an instrument to quantify them, so that the interpretive judgement about authorship is made on explicit evidential grounds rather than on received scholarly conviction.

Methods

With these pitfalls in view, the two methods below are designed to relocate the evidential anchor from an internal Pauline baseline to an external empirical one. Stylistic indicators vary by almost any conceivable stratum, including time, language, length, and genre. It is impossible to control every potential variable, but we can partially account for these four by studying every available corpus that scholars regard as univocal among authors writing in Koine Greek within Paul's general time period, delimited here as 100 BC to AD 100.

We use "univocality" in an operational rather than an ontological sense. A corpus is univocal, for our purposes, if scholarship consensually attributes it to a single named author, irrespective of what the historical facts about secretarial participation or scribal intervention may have been. The operational sense is needed because the ontological sense, that only one person ever contributed to a piece, is rarely recoverable for high-register compositions in antiquity, since secretaries were commonly used [39] and scribes may have made changes over time to documents. [40] Our term therefore tracks what pieces scholars are content to label under a single author's name. The definition is malleable, proposed in view of the limitations of soundly proving single authorship in ancient texts. Two further assumptions in the methodology are currently unprovable:

1. We assume that scholars can reasonably identify authentic texts from historical figures.
2. We assume that there are concrete stylistic differences between all authors that can be computationally measured.

The result of these limitations is that our method analyses the plausibility of univocality in the Pauline corpus in relation to contemporaneous corpora. Our question is thus: compared to other corpora scholars regard as univocal, is Paul's corpus univocal?

Method 1

Feature selection remains an issue even when every available contemporaneous text is brought in. What are the features that define style? To avoid the issues of high-feature-dimensionality inaccuracies (such as those described in Rosa),[41] low-feature-dimensionality inaccuracies (such as those in White),[42] or biased feature selection (such as those in Pracht and McCauley),[43] we conducted a computational feature-discovery study. Since we cannot know *a priori* which features define these authors' styles, we define style as the cumulative features which hold unique values among all study-group authors. Identifying features with unique values among all authors produces a concentrated feature set that is verifiably useful for authorial discrimination. The method identifies corpus-wide discriminators to answer the question: is this corpus from a distinct author?

Once these features are discovered, authorial ranges are calculated for all authors with a multi-text corpus. Plato's corpus is then introduced to test whether the discriminating features are robust against a held-out author. Paul's variation among the identified corpus-wide features is then calculated to see whether it falls within the range of the other univocal corpora; if so, the Pauline corpus can plausibly be considered stylometrically univocal.

Plato is reserved as an external held-out author for principled reasons rather than admitted into the study cohort. His corpus is chronologically displaced from the 100 BC – AD 100 window and his principal genre (philosophical dialogue) overlaps with Philodemus and, more loosely, with Musonius Rufus among the study authors. The combination of period displacement and partial genre overlap makes Plato a reasonable test of whether the discovered discriminators are picking up authorial signals rather than corpus-specific or period-specific artefacts. Plato's texts were taken from the same source as the rest of the study cohort (the Computing Antiquity database) and processed through the same CLTK lemmatisation and feature-extraction pipeline.

Method 1 is useful for identifying discrete authorship in the aggregate, since it measures features corpus-wide. Aggregating texts into a single unit lets us measure, across unrelated productions, what features remain unique between authors. The feature selection has limitations. First, most features will have unique values; it is unlikely for any two authors to have exactly the same non-zero value for a given variable. The study outcome therefore becomes dependent on the tooling (such as CLTK) and on implementation. Second, while it is useful to identify intra-author feature variation, authors do not produce works in the aggregate but one at a time. Results become contingent on extant productions rather than identifying some true objective ranges. Genre in particular can impact stylometric markers, so that mixed-genre corpora will have enlarged feature variance, and it is not straightforward to control for this when the method is applied to Paul's corpus.

Method 2

Method 2 supplements the previous approach by addressing the issues of corpus-wide feature selection. It verifies whether any of the discovered features can correctly identify the authorship of individual texts. The same preprocessing and feature selection is conducted, but the program also measures features on texts individually. If every text's feature range is closest to its author's aggregated value, then the program can correctly attribute the text to its author. Where it cannot, the feature fails to perfectly assign individual texts and lacks attributive power in the strict sense.

To ensure that the outcome was not limited to a single model, the pipeline ran the analysis under twelve models: "pranaydeeps/Ancient-Greek-BERT", "bowphs/GreBerta",

"cltk_odycy", "stanza_grc_proiel", "lgessler/microbert-ancient-greek-m", "bowphs/PhilBerta", "wantuta/roberta_classifier_menander2", "nlpaueb/bert-base-greek-uncased-v1", "lgessler/microbert-ancient-greek-mx", "bowphs/SPhilBerta", "lgessler/greek_mpt", and "lgessler/microbert-ancient-greek-mxp". Other models, such as "pnadel/llama38B_grc2eng_alignment", were candidates but failed due to memory limitations on the executing host.

If no single feature could perfectly attribute all texts, the program searched for feature combinations of sizes between 2 and 25, restricted by the virtual machine memory size. If no perfect feature groups were identified, the program selected any feature that correctly assigns more than half of all texts. Surviving features were measured against Plato to ensure there are unique values, and the variance of selected features was measured against Paul for every model.

In the multi-model analysis the embedding models lack native lemmatisers, and the pipeline purposely avoids injecting an external one, so as to preserve the distinction between the CLTK pipeline and the embedding-based one and to see what the embedding-model tokenizers produce on their own. The multi-model analysis was subsequently repeated with CLTK lemmatisation applied to the Paul-side scripts, in order to test the robustness of the results once preprocessing is homogenised across the pipeline. Both runs are reported below. The method provides insight into Paul's variance on features that carry attributive power, complementing the corpus-level discrimination of Method 1.

Data

All texts were acquired from version 1.0.2 of the Computing Antiquity database.[44] The database was filtered to include all Greek texts from 100 BC to AD 100. Each text was downloaded in the format "Cleaned, joint text without line separators." No additional source-text manipulation was conducted. The study authors removed pseudepigrapha, such as *The Battle of Mice and Frogs* as a pseudo-Homeric text, and organised each text file into directories assigned to its author.

This process produced texts from thirty-six authors: Adamantius, Aelius Theon, Anubio, Aristonicus of Alexandria, Arius Didymus, Asclepiodotus Tacticus, Bion of Phlossa, Brutus, Cebes, Chariton, Chion of Heraclea, Clement of Rome, Comarius, Crateuas, Didymus Chalcenterus, Dionysius of Halicarnassus, Dorotheus of Sidon, Flavius Josephus, Geminus, Heron of Alexandria, Isidorus of Charax, Lesbonax, Longinus, Mithridates IV of Pontus, Musonius Rufus, Onasander, Parthenius of Nicaea, Pedianus Discorides, Philo of Alexandria, Philodemus, Polemon of Laodicea, Scymnus, Strabo, Theodosius of Bithynia, Tryphon, and Xenophon of Ephesus.

Authors were subsequently removed if they had fewer than 1,000 words, given the difficulty of parsing reliable discriminators from short texts.[45] This threshold removed Anubio, Crateuas, Dorotheus of Sidon, and Mithridates IV of Pontus, retaining 32 authors for the analysis. For feature-variation comparisons, authors with only one extant text were also excluded, leaving a comparison cohort of twelve multi-text authors: Adamantius, Bion of Phlossa, Chariton, Clement of Rome, Dionysius of Halicarnassus, Flavius Josephus, Heron of Alexandria, Lesbonax, Musonius Rufus, Pedianus Discorides, Philo of Alexandria, and

Philodemus. The full retained corpus comprises 2,119,096 words across the 32 multi-word authors.

The Pauline corpus for this study consists of the fourteen letters transmitted under Paul's name in the New Testament canon: Romans, 1–2 Corinthians, Galatians, Ephesians, Philippians, Colossians, 1–2 Thessalonians, 1–2 Timothy, Titus, Philemon, and Hebrews. The Greek text is the Society of Biblical Literature's Critical Edition of the Greek New Testament (SBLGNT). Robustness to the exclusion of Hebrews is reported separately in Results. Paul's corpus is treated as one of the 32 authors in the study cohort, with the same 1,000-word threshold and preprocessing pipeline applied as for the other eleven multi-text corpora.

Implementation

The analysis was run on a virtual machine in Google Cloud Platform (GCP). The virtual machine used the "debian-12-bookworm-v20251111" boot disk source image with 100 GB of balanced persistent-disk storage, 12 vCPUs, and 64 GB of memory. Twelve vCPUs and 64 GB of memory were the maximum available at the study author's GCP tier, limiting our ability to run large models which consume high amounts of memory during processing.

Programmatically, features were first extracted from all texts using Python's Classical Language Toolkit (CLTK) for ancient Greek (grc). After initialisation, and after filtering texts according to the 1,000-word threshold, the program iterated through the author directories to process each text. Texts were cleaned with NFD normalisation for word decomposition. Greek letters (U+0370–U+03FF, U+1F00–U+1FFF), combining marks (U+0300–U+036F), and whitespace were retained. Whitespace was normalised and edges were stripped. NFC normalisation then recomposed the characters for CLTK.

CLTK was used to convert words to lemmas. If a text exceeded 900,000 characters, the program split it into chunks of 100,000 words for processing. If lemmatisation failed for a particular word, the original surface form was used as a fallback. After thresholding, cleaning, and lemmatisation, all author texts were combined for feature extraction.

A wide range of features was measured to reduce selection bias while balancing computational restrictions. Morphological features included particle frequencies, case endings, and verb endings. Phonetic features included vowel ratios and frequencies, consonant-cluster frequencies, and diphthong frequencies. Vocabulary richness was measured through simple TTR, MATTR at window sizes 50, 100, and 200, *hapax* and *dis legomena*, average word frequency, and word concentrations. Statistical word measures such as average, median, standard deviation, and maximum and minimum lengths were captured alongside length distributions. Feature dictionaries, including character n-grams (2-, 3- and 4-character sequences) and word n-grams (2- and 3-word sequences), were computed along with individual word frequencies. All features were tested for discriminative power using a separation score that combines coefficient of variation with a Gini-based gap-uniformity measure.

Features were selected using a minimum normalised gap criterion: the smallest pairwise difference between any two sorted author values must exceed 0.1% of the feature's total range, ensuring separation above random chance for N=32 authors. The

attribution scale floor for single-text authors was empirically calibrated using the median coefficient of variation observed across multi-text authors (≈ 0.34), rather than an arbitrary constant, so that single-text authors were not given implausibly narrow attraction basins in the classifier.

One methodological note on uncertainty belongs here. All percentages reported below are point estimates. No bootstrap, permutation, or parametric confidence intervals are attached to them. The aim of the study is descriptive and exploratory: to characterise the range of observed stylistic variance in a Koine Greek corpus in which the number of surviving texts per author is small, and often one or two, so that sampling-based uncertainty estimation is itself unreliable on the same data. Point estimates suffice for the claim the paper actually advances, which is comparative rather than population-level — that Paul's internal variance sits within the envelope defined by twelve other multi-text corpora across 231 features, not that any of these figures constitutes what Paul's variance "truly is." Future work with larger per-author corpora, or with synthetic resampling designs calibrated to short-text regimes, could supply interval estimates for the quantities discussed here.

Results

Method 1

The program identified 231 features with minimum normalised gap exceeding 0.1% of feature range across all authors, which could correctly separate all agglomerated authorial corpora. The distribution across feature types was: character 2-grams (92), character 3-grams (90), character 4-grams (29), morphological (7), phonetic (5), word frequencies (4), word lengths (3), and vocabulary (1). The top ten features, ranked by separation score (which combines each feature's coefficient of variation with the Gini coefficient of the gaps between sorted author values), were:

Rank	Feature	Type	Separation Score
1	ζεπί	char_4grams	1.541
2	ῥς	char_2grams	1.411
3	ωαῦ	char_3grams	1.365
4	ραν	char_3grams	1.352
5	ζόπα	char_4grams	1.338
6	εἰς	char_3grams	1.328
7	μεθ	char_3grams	1.321
8	ροζμ	char_4grams	1.317
9	θος	char_3grams	1.307
10	ὐδ	char_2grams	1.307

All 231 discriminators survived the Plato robustness test (231/231, 100%), demonstrating that the discriminators generalise as authorial separators across time periods.

Method 1 — Paul's variance against the twelve multi-text authors

Of the 231 features, compared to authors with multi-text corpora, Paul's internal variance was less than Bion of Phlossa for 66.1% of the features and less than Heron of Alexandria for 57.4% of the features. Paul varied more than the remaining ten authors. Paul's average consistency against all twelve authors was 31.9%, meaning the Pauline corpus showed higher variance than the averaged twelve multi-text corpus authors on approximately 68% of features.

Author	Comparisons	Paul Higher %	Consistency
Bion of Phlossa	171	33.9%	66.1%
Heron of Alexandria	216	42.6%	57.4%
Lesbonax	132	53.8%	46.2%
Clement of Rome	205	59.5%	40.5%
Musonius Rufus	204	62.7%	37.3%
Dionysius of Halicarnassus	225	68.4%	31.6%
Pedianus Discorides	216	75.5%	24.5%
Philodemus	195	77.9%	22.1%
Chariton	217	78.8%	21.2%
Philo of Alexandria	225	87.1%	12.9%
Adamantius	203	87.2%	12.8%
Flavius Josephus	224	89.3%	10.7%

Excluding Hebrews from the Pauline corpus produced an average consistency of 32.5% (an improvement of 0.6 percentage points). The aggregate interpretation does not change with the inclusion or exclusion of Hebrews: Hebrews is not the dominant driver of Paul's observed internal variance.

Repeating the Paul-variance analysis after applying identical CLTK lemmatisation to all Paul-side scripts (so that preprocessing is homogenised with Method 1) produced an average consistency of 30.6% (with Hebrews) and 30.4% (without Hebrews). The sign of the Hebrews effect reverses under lemmatisation, consistency decreases from 30.6% to 30.4% on removal rather than increasing as in the non-lemmatised run, but the magnitude of the shift is well under a percentage point in both directions. The top two most-similar authors, Bion of Phlossa and Heron of Alexandria, retain their positions; the lower mid-tier reorders modestly.

The quantitative shift is small, and the qualitative interpretation, that Paul exhibits high but plausibly univocal internal variance, is unchanged.

Method 2 — non-lemmatised pipeline (embedding-native preprocessing)

Every model failed to identify a feature or feature combination that could correctly attribute all texts to their respective authors. Under the empirically calibrated CV floor (≈ 0.34) no embedding dimension passed the 50% attribution threshold, whereas under the earlier 0.05 floor several embedding dimensions (dim_14, dim_34, dim_53, dim_62, dim_74, dim_75, dim_83, dim_90, dim_91) had appeared to cross the threshold in particular models. The adjusted pipeline does not carry them as discriminative.

The baseline for a 50% threshold bears stating. With 32 authors in the analysis cohort, the uniform-prior chance rate for a single-feature classifier is $1/32$, approximately 3.1%, so a feature that attributes half the texts correctly is roughly sixteen times above chance rather than marginally above it. The 50% cut-off was chosen because stricter thresholds returned zero surviving features across every model in the pipeline; the intent is to isolate the strongest single feature each model can propose, not to impose a near-perfect bar which the data do not meet for any architecture.

Eleven of the twelve models identified exactly one surviving feature above 50% attribution accuracy: the character bigram $\varsigma\alpha$ (char_2gram) at 50.5% accuracy (51 of 101 texts, the complete evaluation set, comprising every text from all 32 retained authors, correctly attributed to their author of record). The exception among the eleven was stanza_grc_proiel, which achieved 51.5% (52/101) on the same feature. cltk_odycy identified no features above 50%. The 50.5% aggregate is, however, driven largely by the twenty single-text authors in the cohort, for whom a text-to-author centroid match is near-trivial: on the top feature $\varsigma\alpha$, twenty-two of thirty-two authors score 1.0, almost all of them single-text, while most multi-text authors score well below 0.5 (Heron of Alexandria 0.083, Chariton and Dionysius of Halicarnassus both 0.0). Read against the multi-text cohort alone, $\varsigma\alpha$ does not meaningfully attribute individual texts; its aggregate accuracy is a property of the single-text bloc, not of the feature's discriminative power. The uniformity of this result across architectures, classical NLP pipelines, ancient-Greek BERT variants, GreBerta and its Philological descendants, microbert variants, greek_mpt, and the Menander-trained roberta classifier, establishes that, asked to discover features on its own, the embedding stack converges on a classical surface-pattern statistic and does not propose anything more expressive.

Method 2 — lemmatised pipeline (preprocessing homogenised with Method 1)

Repeating the Method 2 pipeline with CLTK lemmatisation applied to the Paul-side scripts produces qualitatively identical results. Of the thirteen multi-model result files (twelve

per-model plus the aggregate summary), twelve are byte-identical to the non-lemmatised run, including all cltk_odycy, stanza, pranaydeeps, GreBerta, PhilBerta, SPhilBerta, greek_mpt, microbert variants, and nlpaueb bert outputs. The single difference is wantuta/roberta_classifier_menander2, which moves from 51/101 (50.5%) in the non-lemmatised run to 52/101 (51.5%) in the lemmatised run. That sits within the expected noise range for a text-classification pipeline whose output can vary with GPU state and batch ordering, rather than being evidence of a lemmatisation effect on the discriminator selection. The character bigram $\varsigma\alpha$ remains the only feature above 50% accuracy in either run, and no embedding dimension survives the calibrated CV floor under either preprocessing configuration.

In the non-lemmatised multimodel variance analysis the Pauline corpus shows higher variance in 211/276 comparisons (76.4%), but its variance falls within the per-model comparand ranges in every model that uses $\varsigma\alpha$ as its selected discriminator. Paul varies less than Philodemus and Bion of Phlossa, whose corpora consist respectively of philosophical treatises and poetry, a pattern consistent with Method 1.

Discussion

Aggregate interpretation

Under both methods the Pauline corpus shows lower internal variance on key features than Bion of Phlossa, and a method-dependent lower variance than Philodemus and Heron of Alexandria. The Pauline corpus displays high internal stylometric variance compared to contemporaneous multi-text corpora, yet that variance still falls within a plausible range of univocality when measured against the other authorial corpora in our analysis. Hebrews is a useful test of the finding: removing it shifts the aggregate consistency by less than a single percentage point, and the order of authors relative to Paul's variance is preserved. Paul's combined authorial variance, as computed here, is not without precedent; Brown records the same judgment having been reached earlier in the stylometric tradition.[46]

The comparison cohort is generically heterogeneous, and that heterogeneity has to be registered before the aggregate figure is given too much weight. Bion of Phlossa is a poet; Philodemus writes philosophical treatises; Heron of Alexandria produces technical mathematical and mechanical writing; Chariton writes prose fiction; Josephus writes historiography and Philo philosophical theology; Dionysius of Halicarnassus moves between rhetorical criticism and historiography; Clement of Rome writes early Christian epistolary; Adamantius writes Christian dialogue; Pedianus Discorides writes pharmacology; Musonius Rufus produces philosophical discourses; Lesbonax writes rhetorical exercises. Paul writes occasional epistolary. Genre is a known confound for intra-authorial variance, in directions partially independent of authorship: a poet moving across lyric metres will show wider lexical variance than a technical writer on hydraulics even if both are single hands. What the comparison does license is the observation that Paul's internal variance is not uniquely extreme among ancient Greek authors with multi-text corpora; it sits at the variant end of a distribution that already includes deliberately heterogeneous single-author corpora. What it does not license is the claim that Paul's variance is identical to any single comparand's, or that genre has been ruled out as a contributor to that variance. The paper's claim is about the *range* of variance observable across the period, not a genre-controlled identity. A partial structural response to the genre objection is present in the 231-discriminator set: Plato, whose philosophical-dialogue genre overlaps with Philodemus, is held out from the study cohort,

and all 231 features survive the Plato test. That is evidence that the selected features are not simply tracking a genre boundary, although it does not by itself dispose of the general confound.

The aggregate result is compatible with the methodological asymmetry noted in Perry: content-based examinations of authorship tend to skew towards claiming inauthenticity, since discrete thematic agreements are comparatively easy to read as present-or-absent, where style-based arguments pull towards authenticity once the authorial envelope is honestly estimated. Stylometrically, the Pauline corpus comes out as a plausibly single-authored corpus, which is to say, not as a corpus that requires multi-author partitioning before it will yield a coherent stylistic signature.

The single-discriminator result

Method 2 producing only a single discriminator with more than 50% attribution accuracy is surprising at first but understandable given the 32-class single-feature classification criterion. For a single-feature discriminator to achieve 100% attribution accuracy across 32 authors, it needs 32 non-overlapping intervals on a 1-D number line. Empirically, these intervals collapse across authors. No combination of up to twenty-five features survived the attribution threshold either. The result reflects a conceptual step from Method 1 (which measures distinctiveness at the aggregated-corpus level) to Method 2 (which measures it at the individual-text level): features that discriminate corpus-wide do not necessarily possess the resolution required for individual-text attribution when intra-authorial variance is as broad as the data suggest.

This is not necessarily a weakness of the pipeline but a property of the problem. Individual-text attribution in this setting amounts to asking whether each text of each author's corpus sits closer, on a feature axis, to its own author's centroid than to the centroid of any of the thirty-one others. Where the underlying feature shows even moderate intra-authorial variance, the answer will be negative a non-trivial fraction of the time. The interpretive significance of the single-feature result is therefore that discrimination between texts from univocal corpora is extremely difficult to achieve, especially when the analyst is committed to honestly reporting variance within an authorial stylome rather than suppressing it to obtain cleaner classification.

Embedding models and the CV-floor effect

A methodological finding of the Method 2 pipeline deserves direct discussion, since it shifts how the results of similar studies should be read in future. An earlier version of the Method 2 analysis used an attribution scale floor of 0.05 for single-text authors, a constant inherited from prior pipelines. Under that floor several embedding dimensions (dim_14, dim_34, dim_53, dim_62, dim_74, dim_75, dim_83, dim_90, dim_91) appeared to cross the 50% attribution threshold in particular models, suggesting that transformer-based representations carried stylometric signal beyond what classical character-level features could detect. After empirical calibration of the floor to the median coefficient of variation observed across multi-text authors (≈ 0.34 , a roughly seven-fold increase), no

embedding dimension survives. The only feature crossing 50% accuracy across all models is the character bigram $\varsigma\alpha$.

The mechanism is straightforward once the geometry of the classifier is described. The Method 2 classifier attributes a text by computing a standardised distance to each author's mean feature value. The standardisation denominator is the author's observed standard deviation for that feature, taken directly from the data for the twelve multi-text authors, and imputed via the floor for the twenty single-text authors (roughly 62.5% of the authors in the study), for whom the observed standard deviation is zero and the distance would otherwise diverge. The old floor of 0.05 is approximately 6.8-fold tighter than the empirical median coefficient of variation observed across the multi-text authors (0.34), so the imputed denominator for single-text authors was about 6.8-fold smaller than a calibrated floor would make it. Because the denominator enters the distance inversely, the standardised distance from a text to any single-text author's mean was correspondingly about 6.8-fold larger than it would be under the calibrated floor. A text that sits only slightly off a single-text author's mean is therefore assigned a disproportionately large distance penalty against that author, and the classifier's argmin routes the text away from single-text authors and towards multi-text authors. Features that benefit from this routing bias, which are chiefly the embedding dimensions, which carry weak stylometric signals relative to their noise, appear more discriminative than they are, since the cases in which they would be misattributed to a single-text author are systematically suppressed. Calibrating the floor to the empirical median removes the bias, and those features drop below the 50% threshold. The character bigram $\varsigma\alpha$ is unaffected: its between-author differences are structurally large enough that the distance geometry routes texts correctly regardless of scale assumptions.

What this shows is that transformer-based embedding models (Ancient-Greek-BERT, GreBerta, PhilBerta, SPhilBerta, microbert variants, greek_mpt, roberta_classifier_menander2) contributed no useful stylometric signal beyond what raw character-bigram statistics already provided in this study. Across eleven models, every one identified the same classical feature ($\varsigma\alpha$) as the only viable discriminator. The embedding dimensions that had previously appeared useful were artefacts of an overconfident scale assumption about single-text authors. Neural representations are not thereby rendered categorically useless for authorship attribution; they may yet perform better in settings with many more texts per author, larger training corpora, or domain-specific fine-tuning. In the particular setting of ancient Greek with many single-text authors and small corpora, however, the models' learned representations add nothing to the analysis. The discriminative signal resides in the surface-level character patterns, which require no model at all. The pipeline's value, in the end, was not in discovering new features but in empirically revealing classical features.

The lemmatisation comparison

Reporting the Method 2 analysis under both the non-lemmatised embedding-native pipeline and the CLTK-lemmatised pipeline serves two distinct epistemic functions. The non-lemmatised run documents what the embedding models propose as discriminators on their own. The lemmatised run tests the robustness of the substantive result once preprocessing is homogenised with Method 1. The two runs agree: character bigram $\varsigma\alpha$ at 50.5% (or 51.5% in two models), no surviving embedding dimensions, and Pauline variance falling within comparand ranges on the same features. That agreement

is evidence that the qualitative claim of this paper does not depend on a preprocessing choice. Method 1, which already applied lemmatisation in both runs, is byte-identical between them, so the entire Method 1 finding (the 231 discriminators, their separation scores, the 100% Plato robustness) is by construction preprocessing-invariant. The Paul-variance scripts show small quantitative shifts under lemmatisation (consistency 31.9% → 30.6% with Hebrews, 32.5% → 30.4% without), consistent with lemmatisation slightly expanding the overlap of n-gram types between Paul and shorter-corpus comparanda; at the aggregate level these shifts are within a few percentage points and the rank-order of closest-to-Paul authors is stable. We report both figures so that the reader can see both the robustness and the limits of the finding.

Paul, Hebrews, and the Synoptic analogy

The thematic divergence of Hebrews from the remainder of the Pauline corpus is not in serious dispute. The letter opens without the *praescriptio* and *salutatio* that frame every undisputed epistle; its sustained exposition of Christ's high-priesthood has no counterpart in Romans, the Corinthian correspondence, or the Thessalonian letters; and its polished rhetorical periods sit awkwardly beside the more disputatious prose of Galatians. On thematic grounds one would expect Hebrews to pull the aggregate signature of the Pauline corpus in a distinctive direction, or at least to inflate internal variance in ways that a computational procedure would register. Our results do not bear this expectation out. Running the same pipeline with Hebrews excluded shifts Paul's average consistency across the twelve multi-text comparands by only a fraction of a percentage point, a movement well inside the noise margin produced by ordinary preprocessing choices. The letter whose thematic distance from the *Hauptbriefe* is often treated as obvious does not leave a proportionate trace in the aggregated stylometric signal.

This divergence between thematic and statistical distance is not peculiar to Pauline studies. It has a longstanding analogue in the Synoptic problem, and the shape of the analogy is instructive. Matthew, Mark, and Luke are grouped together because they share verbatim material in quantities that scholars consistently interpret as indexing literary dependence, whatever the specific direction of borrowing.[47] John is conventionally set apart; its Christology, discourses, and theological vocabulary read as a different register of early Christian writing. Yet, as Mark Goodacre has recently emphasised, the Fourth Gospel contains passages whose verbal agreement with the Synoptics would, if lifted from their narrative frame, be read as direct textual contact.[48] In Goodacre's account, thematic dissimilarity effectively neutralises the force of these agreements for scholars who have already concluded that John stands apart; the striking similarities are discounted because the surrounding contexts differ. The Paul–Hebrews configuration is analogous: Hebrews reads thematically apart from the rest of the corpus, yet when the text is decomposed into the morphological, phonetic, lexical, and n-gram features that anchor our 231 discriminators, the aggregate signal does not pull away. Goodacre's observation that "every scholar and every student sets their own bar for where coincidence ends and direct contact begins"[49] applies, *mutatis mutandis*, to authorship questions no less than to questions of literary dependence. In both domains, thematic and statistical evidence do not move in lockstep, and the judgement about where one set of signals outweighs the other is seldom made on strictly epistemic grounds.

The same disjunction sits behind a second, older verdict about Paul: that his Greek is low-register, utilitarian, and unliterary. The characterisation has some papyrological warrant and was not unreasonable when the comparanda were limited. Against a broader contemporary sample, its statistical position is weaker than the received verdict implies. Several of the authors in our comparison set, such as Philo of Alexandria, Philodemus, Dionysius of Halicarnassus, and Flavius Josephus, are by any reasonable description not writing in the "ordinary" Koine of the documentary papyri. They are, in varying ways, deliberate prose stylists with literary aspirations. Paul's internal variance nonetheless sits above that of Philo, Josephus, and Adamantius on the majority of discriminating features, and within the same order of magnitude as Dionysius. Were the register assumption underwriting older Pauline scholarship correct as a statistical generalisation, one would expect Paul's corpus-internal feature distributions to cluster more tightly, at the lower end of the spectrum, against authors writing in consciously elevated registers. They do not. The data do not license a wholesale reversal of the register verdict (genre and occasion complicate the comparison) but they shift the burden of proof onto the stylistic low-brow claim in a way that is no longer easy to meet.

A third issue follows. The standard argument against Pauline authorship of Hebrews leans heavily on the absence of the *praescriptio* and *salutatio* that open the undisputed letters.[50] The weight this observation is asked to carry sits awkwardly against the weight carried by features that aggregate across the entire text. A missing or present salutation is, in information-theoretic terms, a binary token; each of our 231 discriminators aggregates thousands of sub-lexical and lexical occurrences. Were the presence of the *praescriptio* inserted into our feature set, it would enter as one discriminator among 231 and would weigh accordingly. Traditional scholarship assigns the opening formula a categorical weight that no other single feature is permitted to exert. The disproportion is a convention of the field rather than a consequence of the statistical evidential structure of the text. A bias-minimising approach does not show that the opening formula is irrelevant (it is clearly a feature of Paul's letters) but that the privileged status it has enjoyed reflects human assumptions rather than a pure unbiased weighting.

Contemporary comparanda

The limits of stylometric inference are best illustrated by cases where the method behaves counter-intuitively against well-understood ground truth. Four recent studies, drawn from quite different corpora, converge on the same cautionary point: binary pronouncements about authorship are difficult to sustain once the confounds of persona, collaboration, genre and probabilistic reasoning are given their due.

Fernando Pessoa is the most striking case. Pessoa composed under a series of literary personae (Alberto Caeiro, Ricardo Reis, Álvaro de Campos, and Bernardo Soares) which he termed *heteronyms* and to which he assigned biographies, philosophies, and even metrical habits of their own. When subjected to Burrows's Delta and its derivatives, the heteronyms do not cluster with texts signed in Pessoa's own name; they resolve as distinct authorial signals, exhibiting the properties of individual hands from a purely stylometric point of view.[51] An Iberian-language treatment of the same material reaches a convergent conclusion through partly overlapping methods.[52] The implication is uncomfortable for anyone wishing to read stylometric separation as direct evidence of separate authorship. A single hand, consciously modulating register, vocabulary and syntactic habit, can generate as much inter-class variance

as exists between unrelated writers. Stylometric discordance, in other words, is necessary but not sufficient for inauthenticity verdict.

If Pessoa attacks the inference from one end, the Hildegard of Bingen case attacks it from the other. Hildegard's late visionary works were transmitted through the collaborative intervention of Guibert of Gembloux, whose role as amanuensis-editor was extensive enough that Delta-style procedures assign the disputed texts to Guibert rather than to Hildegard, even though her other secretaries (whose interventions were confined to grammatical tidying) leave her signal intact.[53] The authors frame this under what they term the Synergy Hypothesis: collaboratively produced texts may bear a stylome that is neither that of the primary author nor that of the secretary, but an emergent third signal proper to the collaboration itself. This matters directly for Pauline studies. The use of secretaries in the production of Paul's letters is not a speculative reconstruction but the consensus position of Richards and those who have followed him,[54] and is the hypothesis marshalled by van Nes, Pao, and others to explain the linguistic peculiarities of the Pastorals without recourse to outright pseudepigraphy. If Guibert's editorial hand can overwrite Hildegard's detectable signature, there is no principled reason to expect Tertius, Sosthenes, or any of Paul's other named co-workers to leave the Pauline stylome untouched. A "more varied" Paul, under these conditions, is what the documentary record predicts, not what it excludes.

The Ovidian case supplies complementary points about genre and career phase. Recent computational work on the *Heroides* and the disputed fifteenth letter places the *Epistula Sapphus* within the single-letter group, consistent with Ovidian authorship, and documents that the *Amores* exhibit much greater internal stylistic diversity than the rest of the Ovidian corpus: "nothing in this analysis suggests that *Am.* 3.5 is *not* Ovidian ... but the *Amores* show much more stylistic diversity than Ovid's other works".[55] A single author, writing in a single extant language within a single attested career, can generate one book whose internal variance approaches or exceeds the interval ordinarily read as cross-authorial. The mechanism is mundane, elegiac experimentation, generic play, youthful composition, but its consequences for stylometric inference are not. The *Amores* sit within the same stylome as the *Metamorphoses* only once one accepts that intra-authorial variance is not a constant to be tuned away but a datum of the author's production.

The fourth point of comparison comes from the rapidly growing literature on AI text detection. A recent contribution argues that perfect detection of AI-generated text is not merely technically hard but theoretically bounded: the closer a model's output approaches human prose, the more detection resolves into a probabilistic statement about the likelihood that a given passage departs from an expected distribution, rather than a binary verdict.[56] The architecture of that argument is of a piece with Perry's treatment of the Seventh Epistle, where the outcome is framed as a graded assessment of authenticity rather than an up-or-down declaration. The contrast with Koentges, who concludes on the basis of a stylometric majority that the *Menexenus* "was not written by Plato", is instructive. The two approaches are not merely differences in method but differences in what the methodologist is willing to report. A probabilistic frame reports intervals and likelihoods; a binary frame reports verdicts, and in doing so tends to under-represent the uncertainty that the underlying measurements actually carry.

These four cases converge on a cumulative argument. Pessoa establishes that one hand can produce several detectable stylomes. Hildegard and Guibert establish that collaborative production leaves a fingerprint of its own rather than an average of its contributors. Ovid establishes that genre and career phase can enlarge intra-authorial variance to a degree which a naïve cross-author comparison would misread. The AI-detection

literature, and Perry before it, establishes that the honest output of such analyses is a probability, not a verdict. The Pauline corpus is the confluence of all four complications: it contains a small number of generically heterogeneous texts, produced with the acknowledged involvement of secretaries, whose authenticity is itself the contested object of inquiry. Under these conditions, the binary question, is this letter by Paul, is the wrong one to ask of a stylometric instrument. What our two methods report is the probabilistic form of the question: is Paul's observed internal variance plausible against the variances of other corpora that scholarship is content to call univocal? Our data indicates that the answer is yes.

Content- versus style-based modes of argument

The foregoing bears on the long-running distinction between content-based and stylometric modes of argument in authorship attribution. Content-based arguments, which compare themes, topics, and discrete lexical agreements, naturally tend to produce verdicts of inauthenticity: it is easy for a motivated forger to reproduce the surface agreements of an author's known positions, and easy for a skeptical reader to find thematic departures that look disqualifying. Style-based arguments track the subconscious habits embedded across a corpus and tend in the opposite direction, because such habits are much harder to forge and much harder to dismiss. The methodology advanced here sits on the stylometric side of that divide by design. Admitting all available features, deriving discriminating power probabilistically, and measuring Paul's variance against a contemporaneous reference class rather than against an *a priori* authentic sub-corpus lets us avoid the inauthenticity-asymmetry that has shaped much of the field from Baur onwards.

Limitations

The preceding argument has been pitched at the level of what the methodology can license; several constraints on its execution bear explicit statement. Text length and text count per author are not controlled. Paul's corpus contains fourteen letters of quite different scale, and the comparison cohort ranges from single-text authors to Philo of Alexandria with over two hundred retained texts. Length-dependent variance estimators behave differently across corpora of such heterogeneous size, and a length-controlled reanalysis is outside the scope of this paper.

The transformer models deployed in Method 2 (Ancient-Greek-BERT, GreBerta, PhilBerta, and the remainder of the battery) were not fine-tuned for authorship attribution. The null result concerns off-the-shelf frozen representations assessed under a feature-discovery protocol; it does not establish that neural representations in general lack authorship signal. A fine-tuned or task-specific pipeline may yet recover discriminative structure that this study does not see.

The univocality assumption for the comparison cohort is itself a scholarly attribution rather than a verified historical fact. The circularity identified in Pauline stylometry recurs one layer up in the selection of corpora that scholarship treats as univocal. The honest response is to mark this limitation and acknowledge that our comparison baseline is no more probabilistically grounded than the scholarly consensus on its members happens to be.

211 of the 231 discriminators are character n-grams (92 bigrams, 90 trigrams, and 29 four-grams). The choice of a wide character-level feature space is defended here on grounds

of surface-pattern uniformity and bias-minimisation, but readers working in the tradition of White or Mealand may prefer higher-level morphological or syntactic features. Whether those features would yield a different set of discriminators is not tested in the present study.

Twenty of the thirty-two authors in the study cohort (62.5%) have only a single retained text above the 1,000-word threshold. The cohort is therefore dominated by authors whose intra-corpus variance cannot be estimated directly, which is what motivated the CV-floor calibration described above. A larger cohort with more texts per author would strengthen all of the variance-based comparisons, and the sensitivity of Method 2 to floor calibration in the present study is a direct consequence of this cohort composition rather than of the method itself.

Conclusion

The lasting contribution of this paper is methodological. Pauline stylometry has been constrained by a circular dependence in which an authentic baseline is assumed, features are curated from that baseline, and results are judged against the same baseline. Our method relocates the circle by replacing the self-referential reference frame with an external one. Paul's variance is measured against the empirical variances of other univocal corpora of the period rather than against a presumed-authentic sub-corpus of his own letters. Features are discovered from their distributional behaviour across thirty-two authors rather than selected *a priori*. The Pauline claim is thereby made falsifiable against data drawn from outside the Pauline corpus. The circularity is not solved in principle but it is replaced by a reference frame that external evidence can adjudicate.

The two load-bearing findings are: 1) Paul's internal variance sits within, though at the variant end of, the empirical envelope defined by the twelve multi-text comparanda; and 2) the character bigram $\varsigma\alpha$ is the sole Method 2 survivor above the 50%-attribution threshold, with no transformer-derived embedding dimension passing empirical calibration of the scale floor.

What the result licenses is support of the framing of Pauline stylometrics as comparative-probabilistic rather than binary-verdict. The appropriate output of such analyses is an interval and a likelihood, not a yes-or-no judgement on a given letter. What the result does not license is any definitive pronouncement about the authenticity of any single Pauline epistle. The question White has recently posed about where Paul sits on the continuum of stylistic consistency is the right frame, and our data place him within the empirical envelope of Koine Greek authorship between 100 BC and AD 100, at the variant end of that envelope but inside it. No binary verdict follows from this placement, and none should be claimed on the evidence offered here.

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Citations

- [1]: White, *Counting Paul*.
- [2]: Pracht and McCauley, "Paul's Style and the Problem of the Pastoral Letters," 9.
- [3]: Tanur et al., *Statistics*, 121.
- [4]: Morton, *The Authorship of the Pauline Epistles*, 7.
- [5]: Morton, *The Authorship of the Pauline Epistles*, 14.
- [6]: Rosa, "Computational Stylometrics and the Pauline corpus," 5.
- [7]: Nasir et al., "An Off-the-shelf Approach to Authorship Attribution," 1.
- [8]: Nasir et al., "An Off-the-shelf Approach to Authorship Attribution," 2.
- [9]: Rosa, "Computational Stylometrics and the Pauline corpus."

- [10]: Sima and Dougherty, "The Peaking Phenomenon in the Presence of Feature-Selection," 1.
- [11]: Holmes, "The Evolution of Stylometry in Humanities Scholarship," 111.
- [12]: Brandwood, "Stylometry and Chronology."
- [13]: Ledger, *Re-counting Plato*, 72.
- [14]: Livesey, *The Letters of Paul in Their Roman Literary Context*, 1–72.
- [15]: Harrison, *The Problem of the Pastoral Epistles*, 18–86.
- [16]: Harrison, *The Problem of the Pastoral Epistles*, 26.
- [17]: For instance, compare Harrison's Diagram VI (ibid., 44) with Diagram V (ibid., 35).
- [18]: Koentges, "The Un-Platonic Menexenus," 212–213.
- [19]: Koentges, "The Un-Platonic Menexenus," 212–213.
- [20]: This introduced another possibility of error: the inputted data had to be correct (see Mealand, "Positional Stylometry Reassessed," 271–275).
- [21]: Morton and McLeman, *Paul, the Man and the Myth*, 91–95; Morton and McLeman, *Christianity in the Computer Age*; Morton, "The Authorship of Greek Prose."
- [22]: Mealand, "Positional Stylometry Reassessed," on δέ specifically at 279–281.
- [23]: The diversity is expressed through the χ^2 statistic.
- [24]: This section concentrates on Morton, primarily because of his influence in the field and because of the extreme position he is willing to take on the inauthenticity of most of the Pauline epistles.
- [25]: Rosa, "Computational Stylometrics and the Pauline corpus," 18–20.
- [26]: Pracht and McCauley, "Paul's Style and the Problem of the Pastoral Letters," 9–14.
- [27]: Brandwood, "Stylometry and Chronology," 85–92.
- [28]: Perry, "Examining the Authenticity of Plato's Epistle VII through Deep Learning," 32.
- [29]: Koentges, "The Un-Platonic Menexenus," 233.
- [30]: Ledger, *Re-counting Plato*, 169, and compare 92–168.
- [31]: Nes, *Pauline Language and the Pastoral Epistles*, 41–68; Pao, *1–2 Timothy, Titus*, introduction; and the discussion in Brown, *An Introduction to the New Testament*, 588.
- [32]: Neumann, *The Authenticity of the Pauline Epistles in the Light of Stylostatistical Analysis*, similarly places the emphasis on the width of the authorial envelope rather than on individual deviations from a putative core.
- [33]: Compare, for example, the application of a bidirectional LSTM network in Platonic studies with a corpus-wide n-gram analysis: the same text can be asked rather different questions by rather different tools, and the answers are not always commensurable.
- [34]: Koentges, "The Un-Platonic Menexenus," 214.
- [35]: Faithlife, "SBLGNT."
- [36]: Ledger, *Re-counting Plato*, 169.
- [37]: Rosa, "Computational Stylometrics and the Pauline corpus," 5.

- [38]: Koentges, "The Un-Platonic Menexenus," 215; Perry, "Examining the Authenticity of Plato's Epistle VII through Deep Learning," 2.
- [39]: Richards, *Paul and First-Century Letter Writing*, 101.
- [40]: Epp, "It's All about Variants," 275.
- [41]: Rosa, "Computational Stylometrics and the Pauline corpus," 19.
- [42]: White, *Counting Paul*, 196–199.
- [43]: Pracht and McCauley, "Paul's Style and the Problem of the Pastoral Letters," 8–9.
- [44]: Mortensen, Beyer, and Fog, "Computing Antiquity."
- [45]: Holmes, Gordon, and Wilson, "A Widow and her Soldier," 406.
- [46]: Brown, *An Introduction to the New Testament*, 588. The question of Paul's authorship of Hebrews is itself a black hole of scholarly disputation but the separation between Paul and Hebrews is clear in Brown's introduction, which treats the (deutero-)Pauline letters separately from other epistles, and is reinforced by Hebrews's lacking of the introductory formula (*praescriptio* and *salutatio* in the *exordium*) and other classical stylistic markers (Koester, *Hebrews*, 42–43 and 46).
- [47]: Goodacre, *The Fourth Synoptic Gospel*, 22.
- [48]: Goodacre, *The Fourth Synoptic Gospel*, 7–8.
- [49]: Goodacre, *The Fourth Synoptic Gospel*, 37.
- [50]: Koester, *Hebrews*, 42–43, 46.
- [51]: Skorinkin and Orekhov, "Hacking stylometry with multiple voices."
- [52]: Skorinkin and Orekhov, "“Reis melhor do que eu.”"
- [53]: Kestemont et. al, "Collaborative authorship in the twelfth century."
- [54]: Richards, *Paul and First-Century Letter Writing*, 101.
- [55]: Nagy, "Some Stylometric Remarks on Ovid's *Heroides* and the *Epistula Sapphus*."
- [56]: Ganie, "Uncertainty in Authorship."